



S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part -1 (Paper-1)

– Hira Kumar

Computer Fundamental

Processing Modes

❖ Process

- A process is a program that is currently executing/ running in a computer.
- When a program is stored on a computer it is called a program, but when it starts running in memory, it becomes a process.

OR

- A process is a program in execution.
- Example –
 - ✓ If you open MS word, then program starts running → it becomes a process.
 - ✓ Again, If you open google chrome, it is another process.

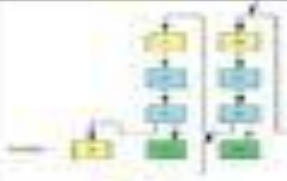
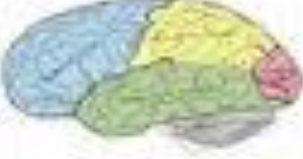
❖ Processing

- The collection and manipulation(तोड़-मरोड़) of data to given Instructions and produce meaningful information." Is called Processing / data processing.
- Example
 - ✓ Calculation marks, sorting a list, updating bank account,....

❖ Steps Of Data Processing OR IPO Cycle – (IPO – Input-process-output)

1. Preparation of source documents
2. Input of data
3. Manipulation of data i.e. Process (shifting, sorting and rearranging the given input)
4. Output of information

5. Storage of information

C T	INPUT	PROCESS	OUTPUT
Computer			
Problem	$2 + 3$		$= 5$
Human			

STAGES OF DATA PROCESSING CYCLE

Input Stage

Data Collection

Data Capture

Encoding

Data Transmission

Data communications

Processing Stage

Performing instructions

Transform raw data into information

Output Stage

Decoding

Presenting data to user

Storage Stage

Storing data

Retrieve data

❖ **Methods of Data Processing**

1. Manual Data Processing

- In manual data processing, data is processed manually without using any machine or tool to get required results.
- Mostly, is processed manually in many small business firms as well as government offices & institutions.
- In an educational institute, for example, marks sheets, fee receipts, and other financial calculations (or transactions) are performed by hand.
- This method of data processing is very slow and errors may occur in the output.

2. Mechanical Data Processing

- In mechanical data processing method, data is processed by using different devices like Calculator, typewriters, mechanical printers or other mechanical devices.
- This method of data processing is faster and more accurate than manual data processing.

3. Electronic Data Processing (EDP)

- Electronic data processing or EDP is the modern technique to process data.
- The data is processed through computer; Data and set of instructions are given to the computer as input and the computer automatically processes the data according to the given set of instructions.
- The computer is also known as electronic data processing machine.

❖ Types of Data Processing

❖ Types of Data Processing by Computer

1. Uniprocessor (Single User System)
2. Batch Processing
3. Multi-Programming
4. Online Processing
5. Real Time Processing
6. Time Sharing / Multi- tasking
7. Distributed Processing
8. Multi-Processing

❖ UniProcessor / Single Processor processing

- Uniprocessor use only one processor (one CPU) but still, it is possible to achieve parallelism by using certain techniques such as pipelining and multitasking.
- Here only one person can work at a time.
- If the system is meant for the use of many persons, much time is wasted waiting for one person to finish his program



➤ Advantages of Uniprocessor

- **Improves performance-** Improves the performance of a uniprocessor by allowing it to execute multiple tasks or instructions simultaneously. This is achieved by increasing

throughput which reduces the time required to complete a particular task.

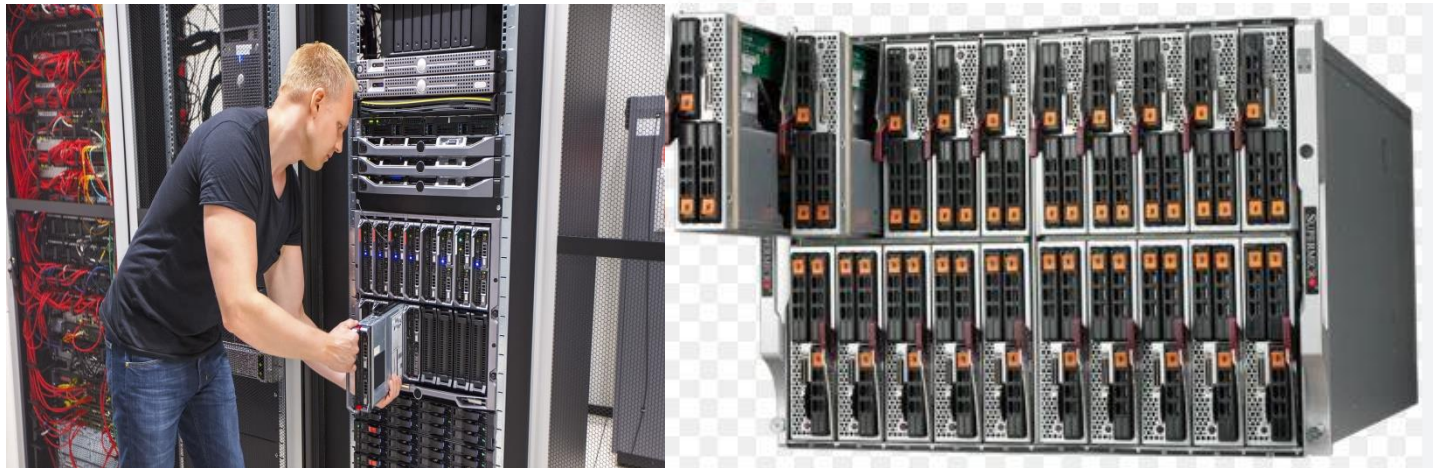
- **Cost Effective-** The cost of a uniprocessor with parallelism is often lower compared to a multiprocessing system.
- **Low power consumption-** A uniprocessor consumes less power than a multiprocessor system which makes it suitable for mobile and battery powered devices.

➤ **Disadvantage of Uniprocessor**

- **Limited scalability**– Parallelism is achieved in a very limited way and as the number of tasks or instructions being executed simultaneously increases the performance decreases. This makes it unsuitable for applications that require high levels of parallelism.
- **Limited processing power**– It has limited processing power as compared to a multiprocessing system hence it is not suitable for applications that require high computational power like scientific simulations and large-scale data processing.
- **Complex design**– Implementing parallelism in a uniprocessor can be complex as it requires careful design and optimization to ensure that the system operates correctly and efficiently this increases the development and maintenance costs of the system.

❖ **MultiProcessor**

- Multiprocessing is the use of two or more [central processing units](#) (CPUs) within a single [computer system](#).
- It is also called GPU.
- The main objective of using a multiprocessor is to boost the system's execution speed.
- There are two types of multiprocessors, one is called shared memory multiprocessor (every CPU uses single memory) and another is distributed memory multiprocessor (every CPU contains own memory).



➤ Advantage

- **Improved performance:** Multiprocessor systems can execute tasks faster than single-processor systems, as the workload can be distributed across multiple processors.
- **Better scalability:** Multiprocessor systems can be scaled more easily than single-processor systems, as additional processors can be added to the system to handle increased workloads.
- **Increased reliability:** Multiprocessor systems can continue to operate even if one processor fails, as the remaining processors can continue to execute tasks.
- **Reduced cost:** Multiprocessor systems can be more cost-effective than building multiple single-processor systems to handle the same workload.

➤ Disadvantages:

Increased complexity: Multiprocessor systems are more complex than single-processor systems, and they require additional hardware, software, and management resources.

Higher power consumption: Multiprocessor systems require more power to operate than single-processor systems, which can increase the cost of operating and maintaining the system.

Difficult programming: Developing software that can effectively utilize multiple processors can be challenging, and it requires specialized programming skills.

Limited performance gains/profit : Not all applications can benefit from multiprocessor systems, and some applications may only see limited performance gains when running on a multiprocessor system.

➤ **Offline Data Entry**

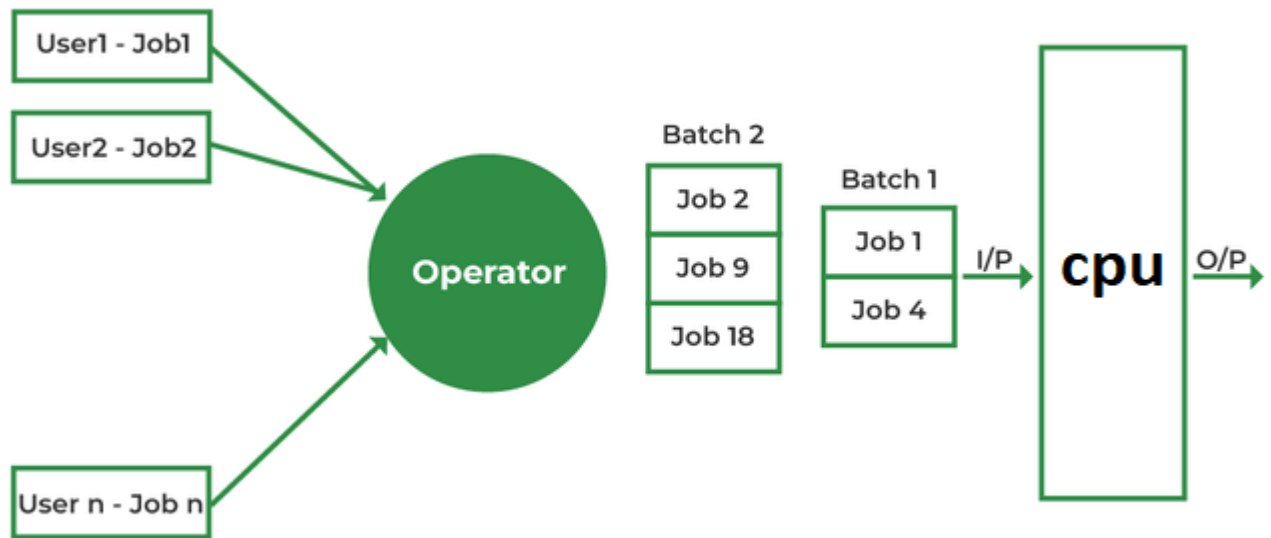
- Putting data from physical or other formats into spreadsheets or any specialized software is a process and it's called offline data entry.
- Converting handwritten forms into a digital database.
- e.g. – Ms Excel, Word,.....

➤ **Online Data Entry**

- Putting information into a computer system in real-time through digital interfaces comes under online data entry process.
- An Internet connection is necessary to perform this process as it directly syncs data to the cloud system.
- e.g. – ATM, Ticket booking,online form filling,...

1. Batch Processing

- Set of jobs/task with similar need.
- No Direct Interaction b/w user & CPU.
- It is an old system but it is still for processing in which the capacity of the C.P.U is limited.
- Here the data is collected and accumulated into batches and processed periodically.



2. Multi-programming

- In batch processing the memory and ALU are not fully utilized.
- Multi-programming enables to store execute more than one program in the C.P.U at the same time.
- In multi-programming is a non-premitive nature.
- Many process (p1,p2,p3,p4,.....) execute one by one. P1 process execute will be until then (तबतक) p1 process not finish. In case p1 need I/O then p2 process execute by CPU. i.e. CPU not keep idle.

3. Multi-Processing

- Multiprocessing systems have 2 or more CPU which have the ability to execute two or more jobs simultaneously.
- Instructions from several programs are carried out by different CPU's at the same time.
- When one CPU fails then the other takes over the processing.

Sr. No.	Multiprocessing	Multiprogramming
1	Multiprocessing refers to processing of multiple processes at same time by multiple CPUs.	Multiprogramming keeps several programs in main memory at the same time and execute them concurrently utilizing single CPU.
2	It utilizes multiple CPUs.	It utilizes single CPU.
3	It permits parallel processing.	Context switching takes place.
4	Less time taken to process the jobs.	More Time taken to process the jobs.
5	It facilitates much efficient utilization of devices of the computer system.	Less efficient than multiprocessing.
6	Usually more expensive.	Such systems are less expensive.

4. Multitasking / Time Sharing Processing

- Multitasking is a technique where it enables a uniprocessor to execute multiple tasks simultaneously.
- This is achieved by dividing the processor's time into short time slots and switching tasks rapidly.
- Each task is given a specific time slot in which the needs to be executed.
- e.g. shopkeeper handel 3 customer simultaneously time slot 10 sec. Teacher handel 10 student simultaneously time slot 30 sec for solve question.

Differences between Multiprogramming and Multitasking operating system

Parameter	Multiprogramming	Multitasking
Definition	Multiple programs are loaded into memory simultaneously.	A single program is loaded into memory and divided into multiple tasks or threads.
Switching	Switching between programs is done by the operating system.	Switching between tasks is done by the operating system or by the program itself.
Algorithm	Programs are scheduled based on their priority or in a round-robin fashion.	Tasks are scheduled based on their priority or using preemptive or non-preemptive scheduling algorithms.
Use case	Used in batch processing and time-sharing systems.	Used in desktop computing and server applications.
Basic advantage	Maximizes resource utilization by minimizing idle time.	Allows multiple tasks to run concurrently to improve system responsiveness.

5. Real Time Processing

- Also called Interactive mode or direct mode
- It is called a real system because it has to give results based on dead lines which are imposed.
- User can have direct contact with the computer through his terminal.
- Data is processed on the spot and no delay between the time data is available and the time it is processed.
- It is the technique of updating files, transactions, data immediately after the event to which it occurs.

BASIS FOR COMPARISON	TIME SHARING OPERATING SYSTEM	REAL-TIME OPERATING SYSTEM
Basic	Emphasis on providing a quick response to a request.	It focuses on accomplishing a computational task before its specified deadline.
Computer resources	Shared between the user.	No sharing takes place and events are external to the system.
Process deals with	More than one application simultaneously.	Single application at a time.
Modification of the program	The programs can be modified and written by the users.	No modification is possible.
Response	The response is generated within the second, but there is no compulsion.	User must get the response within the defined time constraint.

6. Online Processing

- This type of processing is where a user communicates directly with the computer system. → When the user makes a change to the data stored in the computer, the system will automatically update and re-process. This ensures that the system always contains up-to-date information. For example - Music Concert booking systems need to use online processing. This is because the system needs to constantly recalculate how many seats are left as people reserve their tickets.
- Example - Booking systems / Ticket reservations (airlines, cinema, theatre, concerts)
- Bank & ATM's (recalculates how much money is left in the account)
- Purchasing goods on the Internet (e.g. Amazon recalculates how many items are left in stock)

➤ MultiComputers

- Collection of Uniprocessor (More than one computer) is called multiComputers.



	Single Processor Systems	Multiprocessor Systems
	Single processor system contains only one processor for processing.	Multiprocessor system contains two or more processors for processing.
	The performance of uniprocessor is not good	performance of multiprocessor is good
	uniprocessor is less cost	Multiprocessor is more expensive compare to uniprocessor
	speed of the Uniprocessor is Slow	Speed of the multiprocessor is fast
	It is Easy to design Single Processor Systems.	It is difficult to design Multi Processor Systems
	Single processor system is less reliable because failure in one processor will result in failure of the entire system.	Multiprocessor system is more reliable because failure of one processor does not halt the entire system but only speed will be slowed down.
	e.g. Modern PCs / laptop.	Servers

=====HIRA KUMAR=====

